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Master of Computer Applications
Semester – III
Data Analytics for Business
Final Project Report
Title: WorkForce Analytics Dashboard

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Title: Workforce Analytics Dashboard

GitHub Link: https://github.com/lohithslohiths110405-ui/Data-Analytics-for-Business/tree/main/Final_Project

Power BI Service Link: [\[View WorkForce Analytics Dashboard in Power BI Service\]](#)

Problem Statement and Data Collection:

Given Problem Statement:

You are a Business Intelligence Consultant for a growing organization. Your task is to develop an end-to-end Business Analytics solution using a dataset of your choice to support data-driven business decisions.

Project-Specific Problem Statement:

The objective of this project is to develop a Business Intelligence solution for the Human Resources domain using employee-level workforce data. The dashboard analyses employee attrition across department, job role, gender, business travel, overtime, education field, marital status, job satisfaction, job level and tenure at the company. The solution helps an HR analytics team, people-operations manager, or business leadership identify which employee segments are most at risk of attrition, understand the factors associated with turnover, and generate actionable insights to support retention strategy, workforce planning and talent-management decisions.

Data Description:

Dataset used: an employee-level HR attrition dataset (employee_data), hosted on a MySQL database and connected to Power BI via Power Query. Each row represents one employee. Key columns include:

- Age – employee's age in years
- Attrition – whether the employee has left the organization (Yes / No)
- BusinessTravel – frequency of business travel (Travel_Rarely, Travel_Frequently, Non-Travel)
- DailyRate – employee's daily pay rate
- Department – department the employee belongs to (Research & Development, Sales, Human Resources)
- DistanceFromHome – distance between home and workplace
- Education / EducationField – education level and field of study
- Gender – employee's gender
- JobRole – specific job title (e.g., Laboratory Technician, Sales Executive, Manager)
- JobLevel / JobSatisfaction – seniority level and self-reported job-satisfaction score
- MaritalStatus – Single, Married or Divorced
- MonthlyIncome – employee's monthly income
- OverTime – whether the employee regularly works overtime (Yes / No)

- YearsAtCompany – tenure of the employee at the company, in years

Business Problem:

Organizations need a fast way to see which parts of the workforce are actually losing people — not just an overall attrition percentage, but attrition broken down by department, role, tenure, travel demands and overtime load. This project builds a single interactive dashboard that answers that question at both the organization level and the individual-employee level, so HR can prioritise retention effort where it will have the most impact.

Data Cleaning & Transformation (Power Query):

The following steps were then applied in the Power Query Editor before loading the table to the model:

- Connected to the MySQL server and drilled through the Navigator (which listed 286 tables across all schemas on the instance) to isolate the single relevant table, employee_attrition.employee_data.
- Applied a Navigation step to load only that table into a query named MySQL_Employee / EXP_8, discarding every unrelated schema (bank, mysql-system, other demo databases, etc.) returned by the server.
- Verified data types on every column — Age, DailyRate, DistanceFromHome, Education, JobLevel, JobSatisfaction, MonthlyIncome and YearsAtCompany as Whole Number; Attrition, BusinessTravel, Department, EducationField, Gender, JobRole, MaritalStatus and OverTime as Text.
- Checked column-quality indicators (Valid / Error / Empty) for every column — all columns returned 100% Valid, 0% Error and 0% Empty, confirming the source data required no null-handling or de-duplication before loading.

Final_exp

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Export query results Refresh Preview Advanced Editor Choose Remove Keep Remove Split Group Data Type: Whole Number Merge Queries Append Queries Combine Files

Queries [2]

EXP_8 MySQL_Employee

Source = Source([Schema = "employee_attrition", Item = "employee_data"])[Data]

	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education
1	41	Yes	Travel_Rarely	1102	Sales	1	
2	49	No	Travel_Frequently	279	Research & Development	8	
3	37	Yes	Travel_Rarely	1373	Research & Development	2	
4	33	No	Travel_Frequently	1392	Research & Development	3	
5	27	No	Travel_Rarely	591	Research & Development	2	
6	32	No	Travel_Frequently	1005	Research & Development	2	
7	59	No	Travel_Rarely	1324	Research & Development	3	
8	30	No	Travel_Rarely	1358	Research & Development	24	
9	38	No	Travel_Frequently	216	Research & Development	23	
10	36	No	Travel_Rarely	1299	Research & Development	27	
11	35	No	Travel_Rarely	809	Research & Development	16	
12	29	No	Travel_Rarely	153	Research & Development	15	
13	31	No	Travel_Rarely	670	Research & Development	26	
14	34	No	Travel_Rarely	1346	Research & Development	19	
15	28	Yes	Travel_Rarely	103	Research & Development	24	
16	29	No	Travel_Rarely	1389	Research & Development	21	
17	32	No	Travel_Rarely	334	Research & Development	5	
18	22	No	Non-Travel	1123	Research & Development	16	
19	53	No	Travel_Rarely	1219	Sales	2	
20	38	No	Travel_Rarely	371	Research & Development	2	
21	24	No	Non-Travel	673	Research & Development	11	
22	36	Yes	Travel_Rarely	1218	Sales	9	
23	34	No	Travel_Rarely	419	Research & Development	7	
24	21	No	Travel_Rarely	391	Research & Development	15	
25	34	No	Travel_Rarely	690	Research & Development	6	

35 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

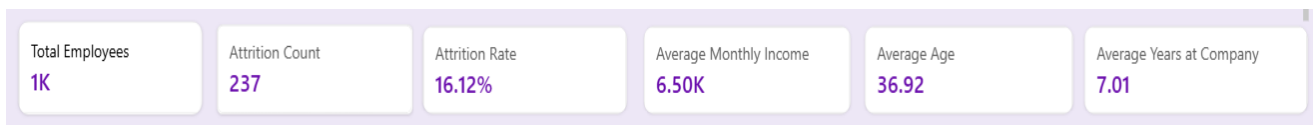
PREVIEW DOWNLOADED AT 10:22 AM

Data Modelling & Relationship:

The report is built on a single, wide fact table rather than a multi-table star schema:

- EXP_8 (fact table) – one row per employee, holding demographic attributes (Age, Gender, MaritalStatus, EducationField), job attributes (Department, JobRole, JobLevel, JobSatisfaction, OverTime, BusinessTravel), compensation (DailyRate, MonthlyIncome) and outcome fields (Attrition, YearsAtCompany).
- Because every attribute needed for the KPIs and visuals already exists on one employee-grain table, no additional dimension or bridge tables were required, and no cross-table relationships needed to be defined in the model.
- All slicers (Department, Business Travel, OverTime, Job Role, Education Field) and visuals filter directly on columns within EXP_8, keeping the model simple and fast to query.

DAX Measures and KPIs:



1.Total Employees

DAX Measure: Total Employees = COUNTROWS('EXP_8')

Insights:

- Represents the total number of employees in the dataset.
- Helps HR understand the overall workforce size.

2. Attrition Count

DAX Measure:

Attrition Count = CALCULATE(COUNT(EXP_8[EmployeeNumber]),EXP_8[Attrition] = "Yes")

Insights:

- Shows the number of employees who have left the organization.
- A higher attrition count indicates a potential employee retention problem that HR should investigate.

3.Attrition Rate

DAX Measure :

Attrition Rate = DIVIDE([Attrition Count],[Total Employees],0)

Insights:

- Shows the percentage of employees who left compared with the total workforce.

- Helps HR evaluate the overall employee retention level and compare attrition across departments or employee groups.

4.Average Monthly Income

DAX measure:

Average Monthly Income = AVERAGE(EXP_8[MonthlyIncome])

Insights:

- Shows the average monthly salary/income of employees.
- Helps HR understand the organization's overall compensation level and investigate whether compensation may be associated with attrition.

5. Average Age:

DAX measure:

Average Age = AVERAGE(EXP_8[Age])

Insights:

- Shows the average age of employees in the organization.
- Helps HR understand the workforce demographic profile and identify whether attrition differs across age groups.

6.Average Years at Company:

DAX measure:

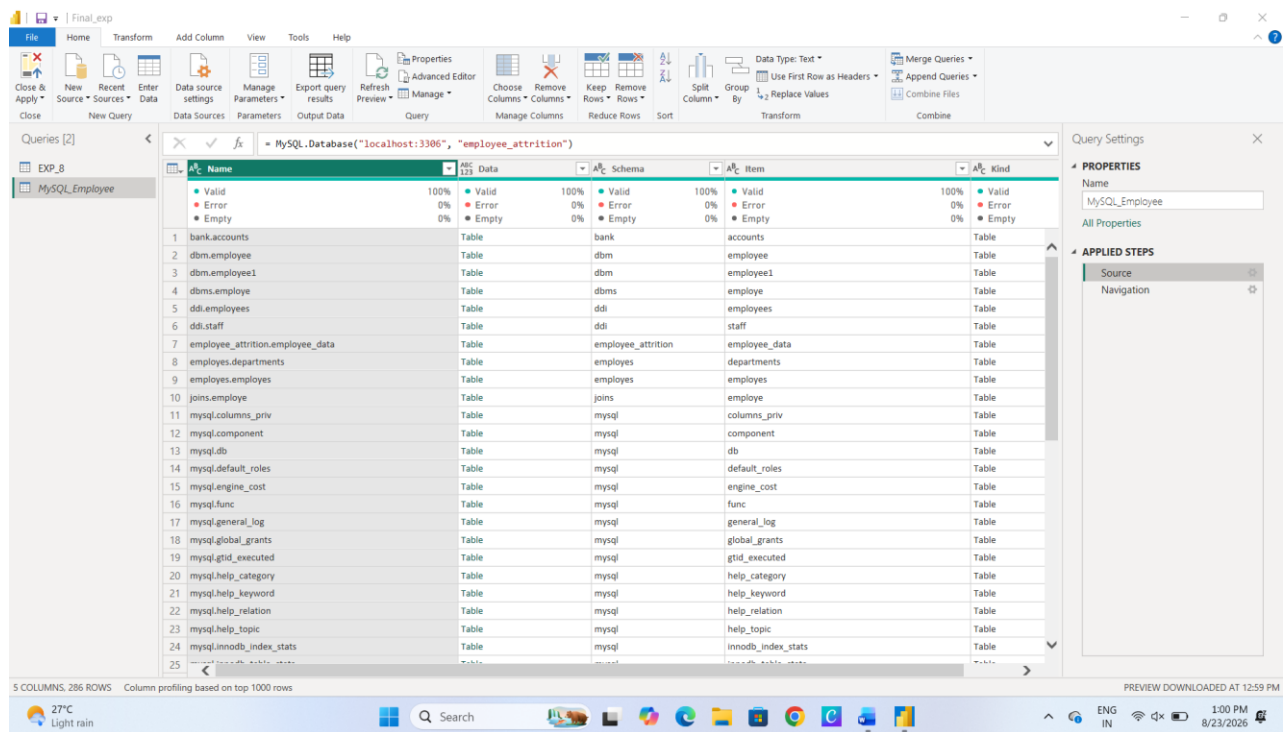
Average Years at Company = AVERAGE(EXP_8[YearsAtCompany])

Insights:

- Shows the average employee tenure in the organization.
- Helps HR understand employee experience and identify whether short-tenure employees are more likely to leave.

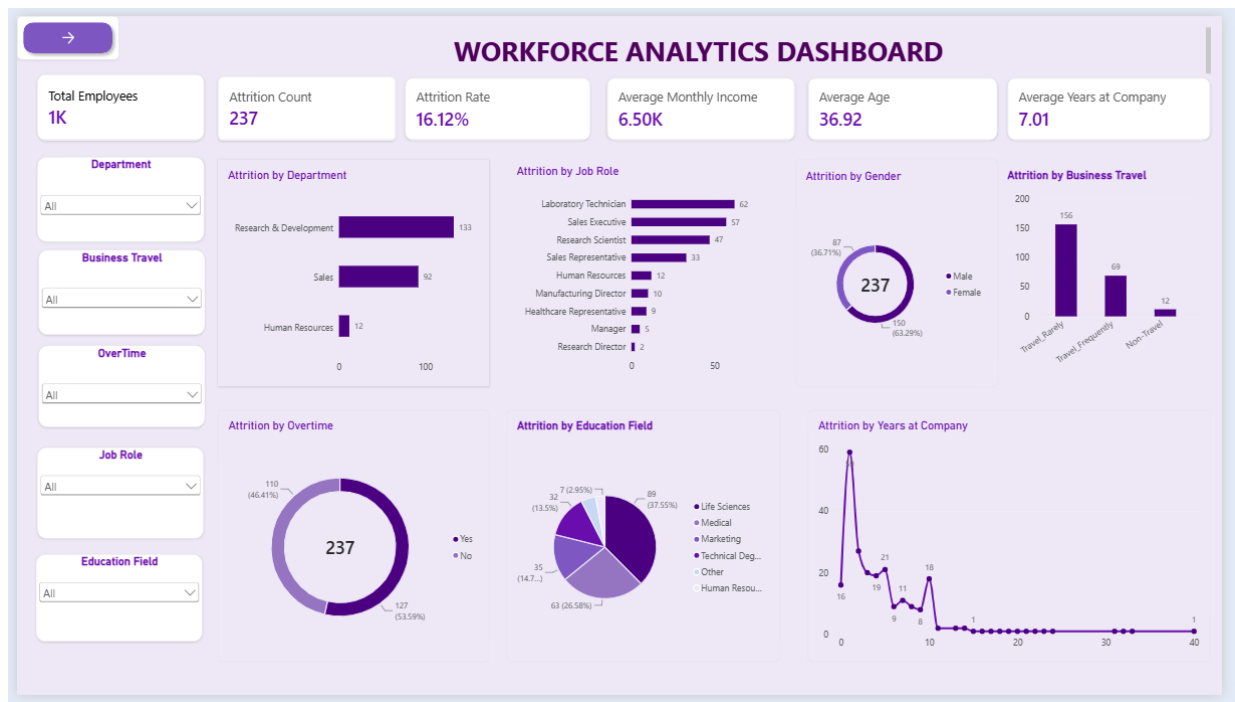
MySQL Server Connection with Power BI Desktop:

- Connected Power BI Desktop directly to a MySQL Server database (localhost:3306) using Get Data → MySQL Database.
- Browsed the Navigator, which surfaced every schema on the instance, and selected the employee_attrition.employee_data table as the source for the model.
- Loaded the table into the Power BI data model with a refreshable connection, so the dashboard can be updated by re-running Refresh once the underlying MySQL table changes.



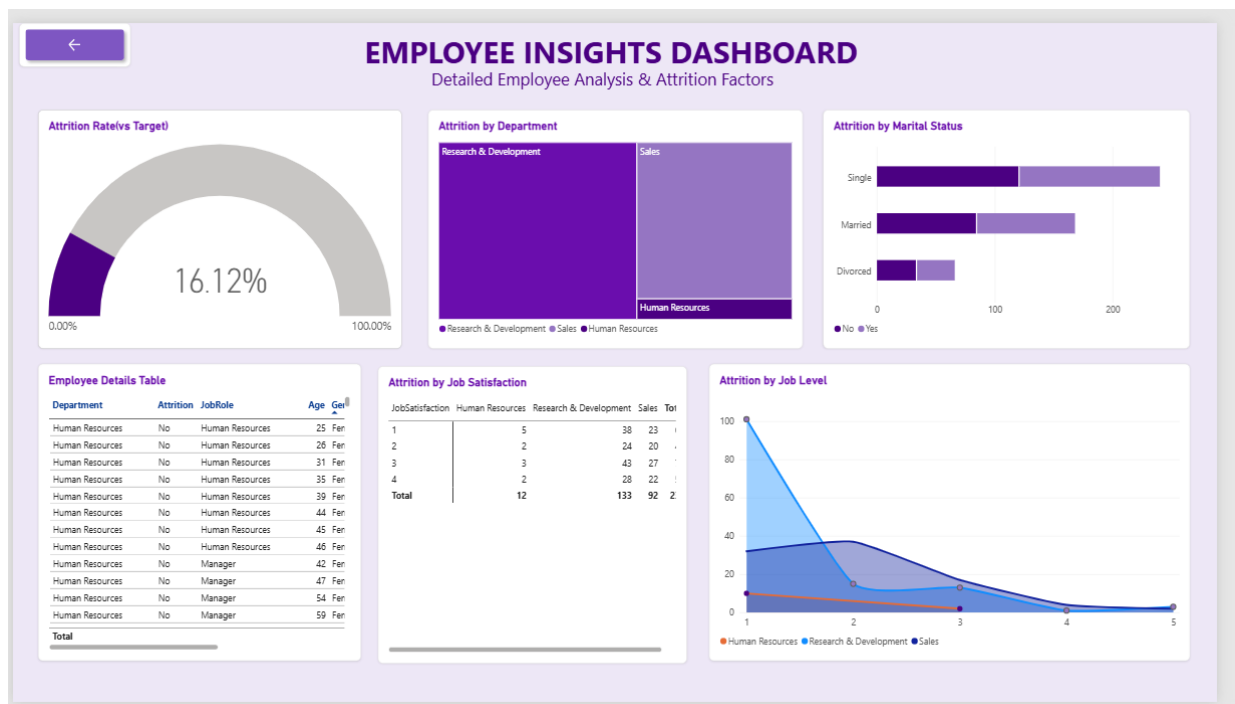
Interactive Workforce Analytics Dashboard Image:

- Workforce Analytics Dashboard provides an interactive overview of employee attrition across the organization.
- Displays key metrics such as Total Employees, Attrition Count, Attrition Rate, Average Monthly Income, Average Age and Average Years at Company.
- Breaks attrition down by Department, Job Role, Gender and Business Travel frequency.
- Shows the split between employees who work overtime and those who do not, and attrition by Education Field.
- Plots attrition against Years at Company to reveal when in an employee's tenure they are most likely to leave.
- Slicers for Department, Business Travel, OverTime, Job Role and Education Field, plus a page-navigation button, let users drill into any employee segment.



A second, drill-through page — the Employee Insights Dashboard — adds further detail:

- An Attrition Rate (vs Target) gauge showing the 16.12% actual rate against a 0–100% target scale.
- A treemap of Attrition by Department, an Attrition by Marital Status bar chart (Single, Married, Divorced), and a scrollable Employee Details Table.
- A pivot table of Attrition by Job Satisfaction (levels 1–4) cross-tabulated by department, and an area chart of Attrition by Job Level across Human Resources, Research & Development and Sales.



Business Question Answered:

I. Which department loses the most employees, and how does it compare with the rest of the organization?

Answer: Research & Development accounts for the largest share of attrition with 133 leavers, followed by Sales with 92 and Human Resources with only 12. Out of roughly 1,470 employees, this means the overall attrition rate sits at 16.12% (237 employees), heavily concentrated in R&D.

II. Which job roles are most exposed to attrition?

Answer: Laboratory Technician (62), Sales Executive (57) and Research Scientist (47) are the three roles with the highest attrition counts, together accounting for well over two-thirds of all leavers, while roles such as Manager (5) and Research Director (2) are far more stable.

III. Does gender show a meaningfully different attrition pattern?

Answer: Male employees make up 63.29% of attrition (150 of 237) versus 36.71% for female employees (87 of 237). Since this roughly mirrors typical workforce composition in the affected departments, gender alone does not appear to be a strong standalone driver of attrition.

IV. How much does business travel frequency relate to attrition?

Answer: Employees who Travel Rarely still account for the largest raw number of leavers (156), but Travel Frequently employees (69) and even Non-Travel employees (12) also churn, suggesting travel frequency should be read alongside other factors like role and department rather than in isolation.

V. Does working overtime increase the likelihood of attrition?

Answer: Yes — 46.41% of attrition cases (110 employees) come from staff who work overtime, a disproportionately high share given overtime workers are a minority of the overall workforce, making OverTime one of the clearest single-factor risk indicators on the dashboard.

VI. Which education fields are most represented among leavers?

Answer: Life Sciences (37.55%, 89 employees) and Medical (26.58%, 63 employees) together make up almost two-thirds of all attrition, consistent with these fields dominating headcount in Research & Development.

VII. At what point in an employee's tenure is attrition most likely?

Answer: Attrition by Years at Company spikes sharply in the first year (60 leavers) then falls off quickly by year 2–3 (19–21 leavers) before tapering to a long, thin tail beyond year 10. This points to onboarding and the first-year experience as the single highest-risk window for retention.

VIII. Is attrition risk relatively even across marital status?

Answer: No — Single employees show the highest attrition, followed by Married employees, with Divorced employees showing the lowest counts, indicating that life-stage/marital status is a relevant segment for targeted retention conversations.

IX. How does job satisfaction relate to attrition once department is factored in?

Answer: The Job Satisfaction pivot table shows attrition present at every satisfaction level (1 through 4) in Research & Development, Sales and Human Resources, with R&D contributing the largest counts at every level — showing that even satisfied employees in R&D still leave, so satisfaction alone does not explain the department's elevated attrition.

X. Does attrition risk fall as employees move up in job level?

Answer: The Attrition by Job Level area chart shows a steep decline from Job Level 1 toward Level 5 across all three departments, with Research & Development consistently the highest line and Human Resources the lowest — confirming that early-career, entry-level employees are the group most in need of retention focus.

Business Insights and Recommendation:

Key Insights:

- Attrition is heavily concentrated in Research & Development (133 of 237 leavers) and, within it, in Laboratory Technician, Research Scientist and Sales Executive roles.
- The first year of employment is by far the highest-risk period for attrition, with a sharp spike at Year 1 that falls away steadily thereafter.
- Overtime is a strong single-factor signal: employees who work overtime make up 46.41% of attrition despite being a minority of the workforce.
- Entry-level employees (lower Job Level) leave far more often than senior employees across every department, with R&D consistently the worst performer at each level.
- Life Sciences and Medical are the two education fields most exposed to attrition, tracking the concentration of attrition in R&D.
- Attrition is not a single-cause problem — it shows up across every job-satisfaction band in the pivot table, meaning satisfaction scores alone will not identify who is likely to leave.

Recommendations:

- Prioritise a first-year retention program (structured onboarding, 30/60/90-day check-ins, mentoring) for Research & Development, since the sharpest attrition spike occurs in an employee's first year.
- Review overtime policy and workload distribution in high-overtime, high-attrition roles such as Laboratory Technician and Sales Executive, since overtime is one of the strongest attrition signals in the data.
- Build targeted career-progression and compensation-review paths for entry-level (Job Level 1–2) employees, who show the steepest attrition drop-off as level increases.

- Investigate role-specific pain points in Research & Development beyond satisfaction scores, since attrition appears across all satisfaction levels there — the driver is more likely tied to role design, workload or career path than reported satisfaction.
- Track Business Travel alongside Department and OverTime rather than as a standalone metric, since frequent travel alone does not explain the bulk of attrition volume.
- Monitor Single and early-tenure employees as a combined at-risk segment for proactive HR outreach and stay interviews.

Conclusion:

This project delivers an end-to-end Business Intelligence solution for the Human Resources domain: employee-level attrition data was connected directly from a MySQL database via Power Query, loaded as a single clean fact table with zero data-quality errors, and summarised with purpose-built DAX measures covering headcount, attrition volume, attrition rate, income, age and tenure. The resulting two-page interactive Power BI dashboard lets a user drill from the full ~1,470-employee organization down to a single department, job role or employee record in a few clicks, answering all ten business questions above and surfacing insights — particularly around first-year attrition, overtime load and entry-level turnover in Research & Development — that can directly inform HR retention strategy and workforce planning.